

PD - Project 1: Dynamic Video to Interactive Curriculum Engine

Domain: EdTech, Video Understanding, NLP, Multimodal AI, Curriculum Generation

Overview

Modern software developers have access to thousands of hours of free educational content on platforms such as YouTube. Despite this abundance of learning material, many learners struggle to convert what they watch into practical engineering skills. A common experience is spending hours following along with tutorials, only to realize later that building a similar project independently is much more difficult than expected. This phenomenon is often referred to as *tutorial hell*.

The goal of this project is to develop an AI-powered learning platform that transforms passive educational videos into interactive, personalized learning experiences. Instead of simply allowing users to watch a tutorial from beginning to end, the system should continuously engage learners by automatically generating practice exercises, quizzes, coding challenges, and checkpoints throughout the video.

The overall objective is not to test whether users can copy code from the instructor, but whether they truly understand the concepts being taught and can apply them in new situations.

Motivation

Watching a programming tutorial is fundamentally different from solving a programming problem.

Most existing tutorials encourage learners to follow instructions line by line. While this creates the impression of learning, very little independent reasoning takes place. When learners eventually start their own projects, they often discover that they remember the syntax but do not understand when or why to apply it.

Educational platforms that provide interactive programming exercises solve this problem effectively, but they require enormous manual effort. Teams of instructors must design curricula, create exercises, prepare automated test cases, and continuously update content whenever technologies change.

This project explores whether artificial intelligence can automate a significant portion of this process.

Rather than creating educational content from scratch, the system should leverage the vast collection of existing technical videos already available online and convert them into structured, interactive learning experiences.

Project Goals

The final system should be capable of taking a technical tutorial video as input and producing an interactive learning session.

Some examples of functionality include:

- understanding the video's transcript
- identifying major concepts and milestones
- recognizing transitions between topics
- generating short concept explanations
- creating coding exercises
- generating multiple-choice quizzes
- producing debugging tasks
- asking conceptual questions before allowing the learner to continue
- tracking learner progress
- adapting future questions based on previous performance

The exact implementation is intentionally left open for interns to explore.

Example Workflow

Although the technical implementation is open-ended, a high-level workflow may resemble the following:

1. A user uploads or provides the link to a technical tutorial video.
2. The system extracts relevant information such as:
 - transcript
 - timestamps
 - visual context
 - code shown on screen
 - slides or diagrams (if present)
3. The AI analyzes the lesson structure.
4. Important learning milestones are identified.
5. At selected checkpoints, the learner is presented with interactive exercises instead of immediately continuing the video.
6. The learner completes the activity.

7. The system evaluates the response and provides feedback.
8. Based on performance, the learner either proceeds or receives additional practice.

Example Scenario

Suppose a tutorial teaches Python dictionaries.

Instead of asking the learner to recreate the instructor's exact example, the system could automatically generate a completely new programming problem requiring dictionary usage.

For example:

Instead of

Create a dictionary storing student names.

the generated exercise might ask

Build a program that stores product inventory using dictionaries and supports searching by product name.

The learner practices the same underlying concept in a different context, promoting genuine understanding.

Potential Extensions

Interns are encouraged to explore additional ideas, such as:

- personalized difficulty adjustment
- prerequisite concept detection
- automatic note generation
- flashcard creation
- spaced repetition
- project recommendations
- concept dependency graphs
- interactive debugging exercises
- code review assistance
- interview-style questions
- collaborative learning modes

Expected Deliverables

Possible project outputs include:

- video ingestion pipeline
- transcript analysis system
- concept segmentation module
- question generation engine
- coding challenge generator
- learner progress tracker
- web application
- evaluation dashboard
- API for curriculum generation

The final architecture is intentionally flexible.

Suggested Open-Source Datasets

The following datasets are well aligned with this project:

Dataset	Why it is useful
HowTo100M	Large collection of instructional videos with transcripts, useful for learning instructional video understanding.
How2 Dataset	Multimodal instructional videos containing subtitles and summaries; useful for transcript analysis and summarization.
Multimodal-Textbook-6.5M	Millions of instructional video keyframes paired with OCR and ASR text for multimodal understanding.
CodeSCAN Dataset	Programming screencast dataset focused on IDE screenshots and OCR, useful for extracting code shown during tutorials.